

Name: _____

Date: _____

Chapter 5 Quiz**Knowledge (10 marks) – Multiple Choice on D2L****Thinking (16 marks)**

1. In a coffee cup calorimeter, a 4.05 g sample of CaCl_2 is dissolved in water. The heat produced raises the temperature of 500.0 mL of water from 23.15°C to 30.30°C . Calculate the molar enthalpy for the dissolution of CaCl_2 . Assume that the density of water is 1.00 g/mL , and the specific heat capacity of water is $4.180 \text{ J/g}^\circ\text{C}$. **USE GRASPI** (5 marks) *exothermic*

$$G: m_{\text{CaCl}_2} = 4.05 \text{ g} \quad c = 4.18 \text{ J/g}^\circ\text{C} \quad q = mc\Delta T$$

$$m_{\text{H}_2\text{O}} = 500 \text{ g}$$

$$T_f = 30.30^\circ\text{C}$$

$$T_i = 23.15^\circ\text{C}$$

$$R: \Delta H_r = ?$$

$$A: q = mc\Delta T \quad \Delta H = nH_r$$

$$q = (500 \text{ g})(4.18 \text{ J/g}^\circ\text{C})(30.30^\circ\text{C} - 23.15^\circ\text{C})$$

$$q = 14943.5 \text{ J} \rightarrow 14.9435 \text{ kJ}$$

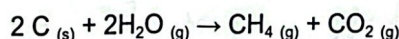
$$q_{\text{system}} = -q_{\text{surroundings}} = \Delta H \quad \Delta H_r = \frac{-14.9435 \text{ kJ}}{0.03649 \text{ mol}}$$

$$\Delta H = nH_r$$

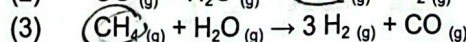
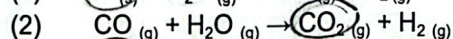
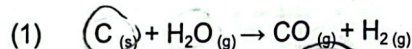
$$n = \frac{4.05 \text{ g}}{110.98 \text{ g/mol}} = 0.03649 \text{ mol}$$

$$= -409.48 \text{ kJ/mol}$$

2. Calculate enthalpy change for the following reaction from the following equations: Show all the steps of your work completely! Make sure to write the target equation. (6 marks)

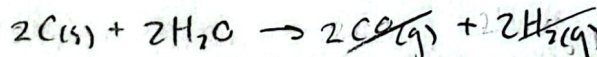


$\Delta H = ?$ the molar enthalpy for the dissolution of CaCl_2 is



$$\Delta H = 206.1 \text{ kJ} - 409 \frac{\text{kJ}}{\text{mol}}$$

$$\checkmark (1) \times 2$$



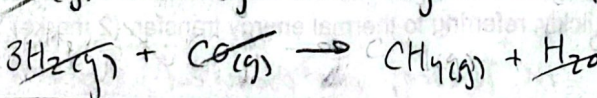
$$\Delta H = 262.6 \text{ kJ}$$

$$\checkmark (2) \times 1$$



$$\Delta H = -41.2 \text{ kJ}$$

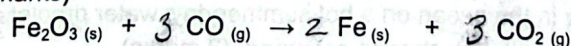
$$\checkmark (3) \times -1$$



$$\Delta H = -206.1 \text{ kJ}$$

$$\Delta H = 97.7 \text{ kJ}$$

3. Use standard enthalpies of formation to calculate the enthalpy change for the production of elemental iron and carbon dioxide gas from solid iron (III)oxide and carbon monoxide gas. **Balance the reaction first!** (5 marks)



$$\Delta H_r = \sum n_{\text{products}} \cdot \Delta H_f^{\circ} \text{ products} - \sum n_{\text{reactants}} \cdot \Delta H_f^{\circ} \text{ reactant}$$

$$= [(2 \text{ mol} \times 0 \frac{\text{kJ}}{\text{mol}}) + (3 \text{ mol} \times -393.5 \frac{\text{kJ}}{\text{mol}})] - [(1 \text{ mol} \times$$

$$-824.2 \frac{\text{kJ}}{\text{mol}}) + (3 \text{ mol} \times -110.5 \frac{\text{kJ}}{\text{mol}})]$$

$$= (-1180.5 \text{ kJ/mol}) - (-492.7 \text{ kJ/mol})$$

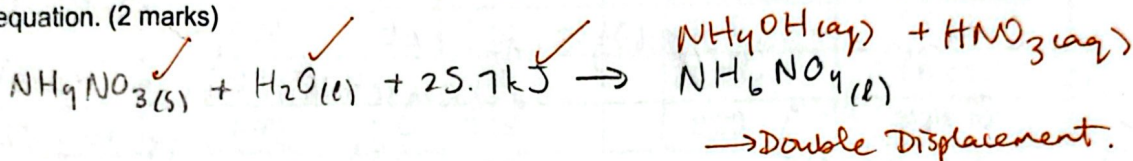
$$= -687.8 \text{ kJ/mol} \rightarrow -24.8 \text{ kJ}$$

Application (10 marks)

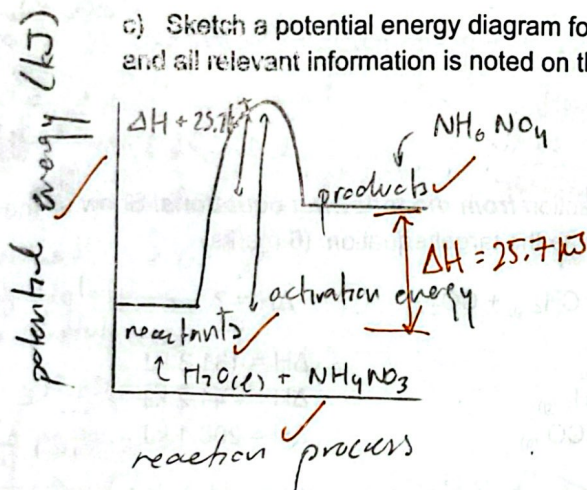
1. A cold pack consists of an inner pouch containing solid ammonium nitrate, $\text{NH}_4\text{NO}_3(\text{s})$, and an outer pouch of water. Twisting the pack breaks the inner pouch and allows the water and ammonium nitrate to mix. As the ammonium nitrate **dissolves**, the temperature of the surrounding environment decreases. The energy change per mol of ammonium nitrate dissociated is +25.7 kJ. (6 marks total)

a) Classify the reaction as endothermic or exothermic. endothermic (1 mark)

b) Write a thermochemical equation for this reaction with the energy term included in the equation. (2 marks)



c) Sketch a potential energy diagram for the reaction. Ensure that all components are labelled, and all relevant information is noted on the graph. (3 marks)



2. Imagine you pour 250 mL of hot cocoa into an insulated Thermos and another 250 mL into a ceramic cup. Explain why the cocoa in the Thermos stays warm for hours but the cocoa in the ceramic cup gets cold quickly, referring to thermal energy transfer. (2 marks)

the thermos is in theory, an isolated system where energy and matter both cannot move in/out. The cup is an open system where they can. The heat (energy) can quickly leave the cup, (cocoa) getting cold while the thermos keeps the heat inside.

3. After swimming in the ocean on a hot summer day, water droplets begin to evaporate from your skin. Explain the enthalpy change occurring. (2 marks)

the water $\text{H}_2\text{O}(\text{l})$ is the system, and the air surrounding it is the surroundings. There is an **Endothermic** reaction happening. Heat is being transferred from the system (H_2O) to the surroundings (air) AKA evaporation. Enthalpy is going system ($\text{H}_2\text{O}(\text{l})$) → surroundings (air).
 water is absorbing heat to evaporate